Eutric Regosols

Description:

- Eutric Regosols are weakly developed mineral soils with limited horizon differentiation, usually found in recent deposits or on young landscapes where soil formation has just begun.
- The term **"Eutric"** indicates **high base saturation (>50%)**, meaning these soils are relatively **rich in basic cations** (Ca^{2+}, Mg^{2+}, K^+, Na^+), which makes them **more fertile and less acidic** compared to Dystric Regosols.
- They typically form on alluvial, colluvial, or aeolian materials, as well as eroded slopes, river terraces, or volcanic deposits. Their weak profile development and high mineral content make them agriculturally useful under good management, though they can be prone to erosion.

Drainage:

- **Wet season:** Generally **well-drained** because of coarse textures and shallow profiles, but drainage can be variable depending on slope and parent material.
- **Dry season:** Soils **dry out quickly** due to low clay and organic matter content. Moisture retention is low.
- **Landscape:** Found on **alluvial fans, terraces, steep slopes, footslopes, and young volcanic plains** with active erosion or deposition.

Depth:

- **Moderately shallow to deep (50–150 cm)**, depending on deposition or erosion rates.
- **Rooting depth:** Usually unrestricted where deep material is present, but may be limited by stones, gravel, or rock fragments in colluvial settings.

Workability:

- **When wet:** Loose and friable; easy to cultivate; not sticky.
- **When dry:** Can become loose and dusty; low cohesion may increase erosion risk under bare conditions.
- **Overall:** Good workability but poor structure and weak aggregation require careful management to maintain productivity.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–15 cmol(+)/kg), depending on texture and organic matter.
- **Base saturation:** High (>50%), due to base-rich parent materials (limestone, basalt, or young alluvium).
- **Nutrients:** Nitrogen and phosphorus are often limiting; calcium, magnesium, and potassium are usually adequate.
- **Organic matter:** Low (<2%) because of rapid decomposition and weak horizon development.
- **Micronutrients:** Generally adequate but may require zinc supplementation for some crops.

pH:
Slightly acidic to neutral (6.0–7.5), depending on the nature of the parent material.

Management:

- Incorporate **organic matter** (manure, compost, crop residues) to enhance aggregation and water-holding capacity.
- Apply **balanced N and P fertilizers** regularly to compensate for nutrient deficiencies.
- Practice **contour farming, cover cropping, or strip cropping** to reduce erosion, especially on slopes.
- Maintain vegetation or mulch cover to prevent crusting and surface sealing.
- Avoid over-cultivation or deep tillage on coarse-textured materials to reduce degradation.
- Suitable for **cereals (maize, sorghum, millet)**, **legumes (beans, cowpeas)**, and **horticultural crops** (tomatoes, onions, cabbages) under irrigation.